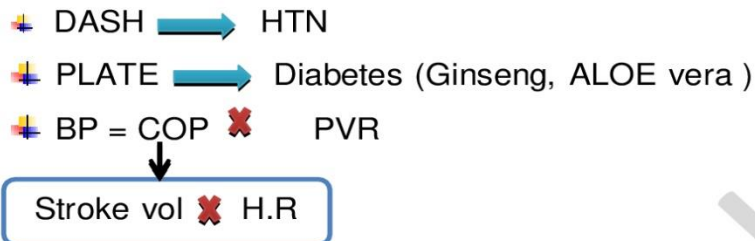


HTN



Diuretics

- A. Thiazide (low ceiling)
- B. Loop Diuretic
- C. K-sparing diuretics



❖ **Thiazide (low ceiling)** Hydrochlorothiazide, Chlorthalidone

✚ **SE:** hypo K, Mg, Na/**metabolic alkalosis** Hyper Ca, glucose, uric acid. "Bone protective"

✚ **CI:** DM, gout, **renal failure**

✚ "thiazide like diuretic" (Indapamide,, clopamide,metolazone)

✚ **Indapamide (vasodilator) which** is used in essential HTN & Pulmonary edema

✚ Metolazone used in renal problem.

❖ **B. Loop Diuretics:** Furosemide, torsemide, **ethacrynic acid**#
diuretic in kidney failure

✚ **SE:** hypo K, Mg, Na, Ca/ **metabolic alkalosis**Hyper glucose, uric acid /

Ototoxicity

✚ **CI:**DM, gout, Aminoglycoside

✚ **Furosemide infusion rate:** 4 mg/min

❖ **C. K-sparing diuretics:** Spironolactone, Eplerenone,
amiloride

✚ **SE:** Hyper K, Gynecomastia, impotence

✚ **CI:**, ACE-I, K supplements, Renal failure

Spironolactone with cimetidine have anti-androgenic effect

✚ **Furosemide : Spironolactone (40 : 100)**

Diuretics

Thiazide

HCZ,,Chlorthalidone
acid

- Pregnancy (())
- "Thiazide like diuretic"
- Indapamide
- Clopamide
- zipamide
- Metolazone???

Loop

Furosemide,torsemide,ethacrynic

→ used in kidney problem

"Ototoxicity"

SE: hypoK, Mg, Na/metabolic alkalosis Hyper, glucose, uric acid.

hibit reabsorption of Na in the ascending limb of the loop of Henle (loops)

or
in the distal tubule (thiazides)

K-sparing diuretics

➔ A. aldosterone antagonist (collecting tube)

- Spironolactone, Eplerenone
- Spironolactone less selective for mineralocorticoid receptor so
Cause..**gynecomastia** and **impotence**..but eplerenone most safe
- ✚ **SE:** Hyper K,,, metabolic acidosis, Gynecomastia, impotence
- ✚ **CI:** ACE-I, K supplements, **Renal failure**

➔ B. Na channel blocker

- ✚ Amiloride,,, triamterene

➤ Carbonic anhydrase inhibitor

Acetazolamide..methazolamide..dorzilamide..brinzolamide

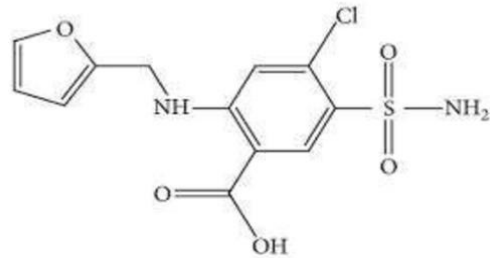
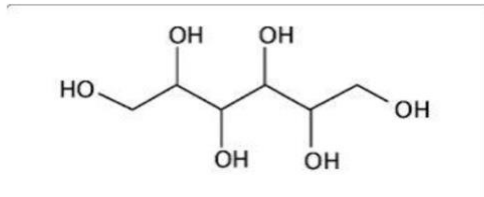
M.A.O,,,, ↓ NaHco₃ reabsorption (block Na/H exchange)

USES, ➔

- Glaucoma
- Urinary alkalization
- Mountain sickness
- Epilepsy

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Mannitol???Osmotic diuretic



◆ Furosemide IV: oral
1: 2

4mg/min

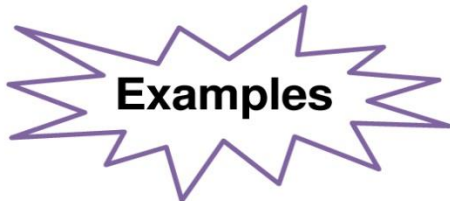
Beta-Blockers:

Tapered off
gradually

non selective
BB C.I in

asthma..D.M

CCB(non
DHP)



- **B1 –blocker**
:Atenolo..Esmolol..Bisoprolol..betaxolol..Metoprolol..Nebivolol
- **BB with HF: Mbc.....**metoprolol..bisoprolol..carvedilol
- **BB with intrinsic sympathomimetic:** acebutalol..pindilol
- **Non selective B + selective α 1 blocker:** labetalol..carvedilol
- **Non selective:** propranolol..sotalol??..timolol..nadolol

Propranolol it is used in	Thyroid storm, HTN, Anxiety, Migraine
Labetalol	use in pregnant with HTN

Less lipophilic: atenolol

High lipophilicity : propranolol

Note

B3 agonist : mirabegron

Antidote : glucagon

B agonist : isoprenalin (non selective)



Glaucoma

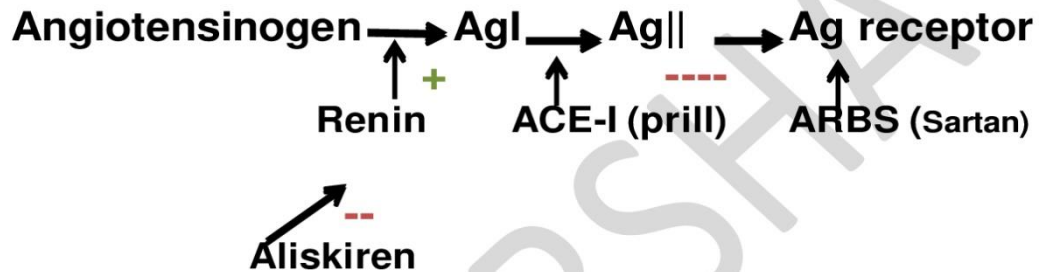
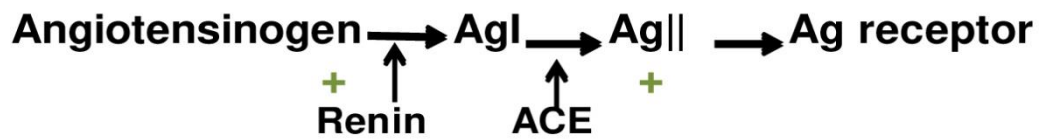
- **Xalatan (latanoprost)**
- **With asthma : latanoprost or betaxolol**
- **With pregnancy :
Brimonidine or timolol**

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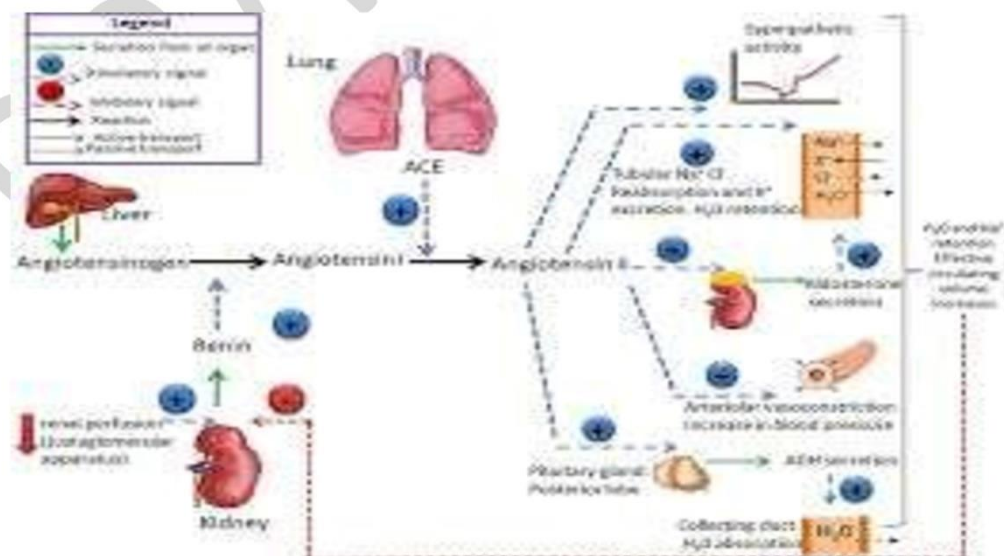
	First trimester	Second trimester	Late pregnancy and lactation
First line of drug	Brimonidine	Brimonidine	Pilocarpine Brinzolamide
Second line of drug	Brinzolamide Timolol maleate	Brinzolamide Timolol maleate PGAs Pilocarpine Oral CAIs	PGAs Timolol maleate Oral CAIs
Avoid	Oral CAIs PGAs Pilocarpine		Brimonidine

*FDA classification of glaucoma medications based on the safety profile of the available drugs: Category B: Alpha-agonists, Category C: Topical beta-blockers, PGAs, topical and oral CAIs, and parasympathomimetics/miotics. # to be used with caution: may rarely cause neonatal electrolyte imbalance and metabolic acidosis; however, acetazolamide is approved by the American Academy of Pediatrics for use during nursing, AGM=Antiglaucoma medications, CAI=carbonic anhydrase inhibitors, PGA=Prostaglandin analogues

Recommended class of AGMs according to the trimester of pregnancy*

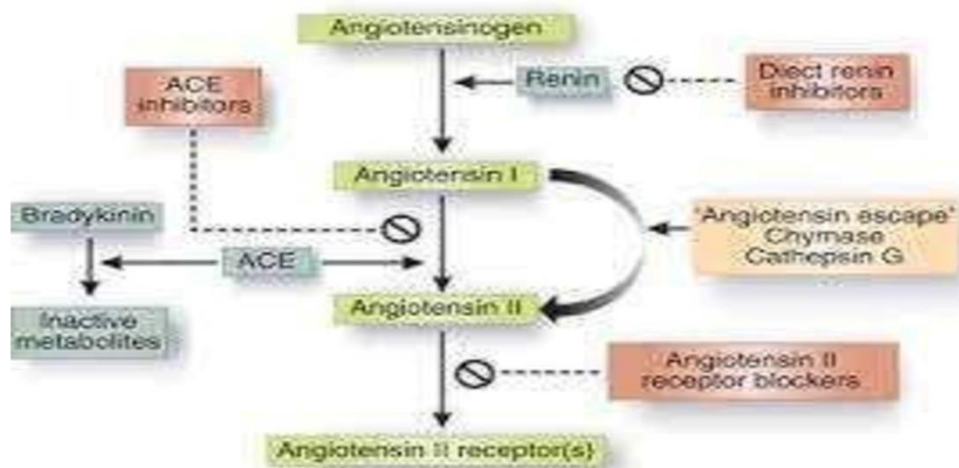


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Difference between : ACEI, ARBS, Renin Blocker and diuretic				
Drug	Renin concentration	Angiotensin 1	Angiotensin2	Angiotensin Receptor
Renin Blocker	↓	↓	↓	
ACEI	↑	↑	↓	
ARBs	↑	↑	↑	↓
Diuretic	↑	↑	↑	

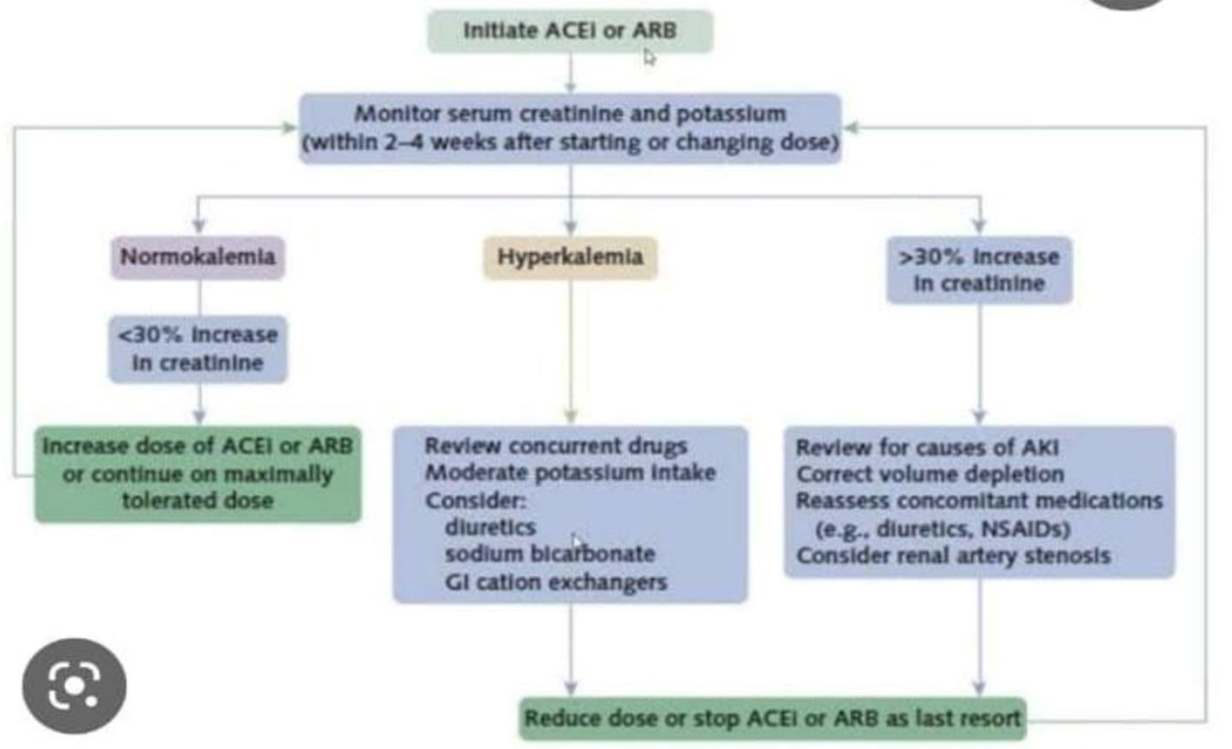
Captopril (sulfer) ... lisinopri.....safe in hepatic

Fosinoprilsafe in renal

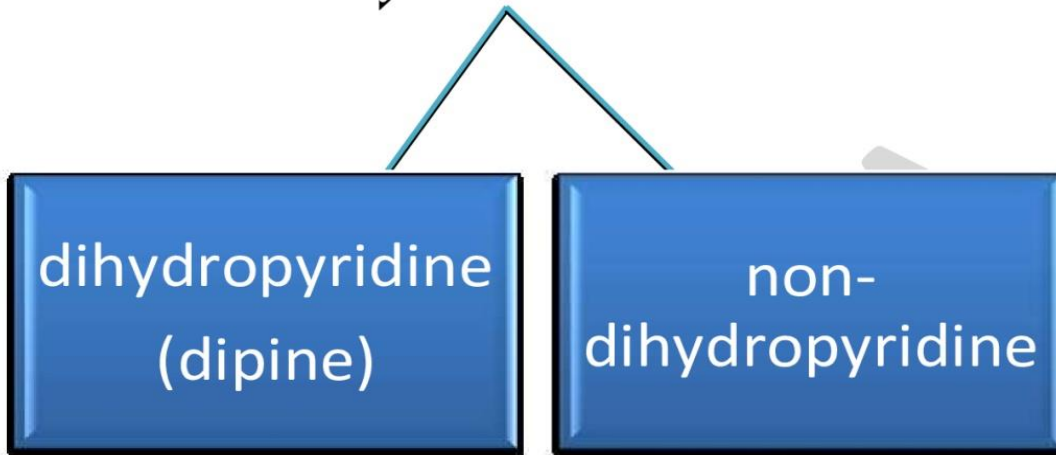
- ✚ ACE-I it is used D.O.C: **prevent** to convert Microalbuminuria to Macroalbuminuria (proteinurea) (D.M)
- ✚ **Contraindication** : angioedema ... cough.. contraindicated in patients with bilateral renal artery stenosis

- ✗ **Angioedema**african american (ethnicity) but use CCB or thiazide
- ✗ **Cough** shift to ARBS
- ✗ **pregnancy**
- ✗ **with bilateral renal artery stenosis**





CCBs ➡

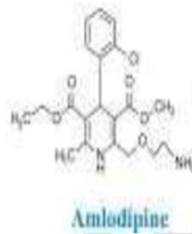
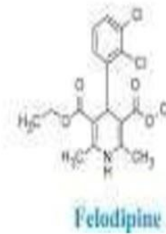
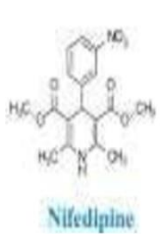
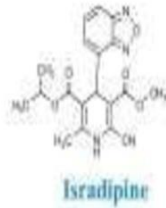
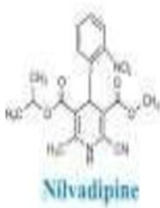


Nifedipin
● act on vascular
S/E ..pripheral edema

verapamil(constipation)..**diltiazem**
● act on heart

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Avoid:

with digoxin??, beta-blockers, Heart block

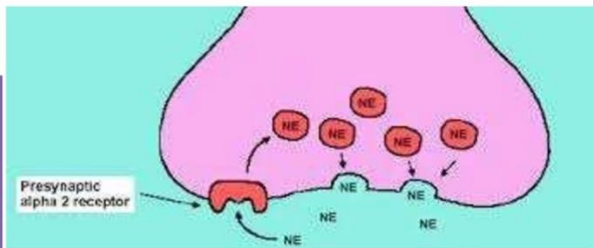
**Nifedipinegingival hyperplasia ?? like what??
..& tocolytic& migrain prophylaxis**



α 2 agonist



CENTRALLY ACTING SYMPATHETIC INHIBITOR



- Clonidine cause hypertension crisis if withdraw suddenly
- α .methyldopa(depression)
- HTN in pregnancy...1. α .methyldopa
2. Nefidipine
3. Labetalol
- Peripherally acting sympathetic inhibitor
 -  Guanethidine
 -  Reserpine
- Selective imidazoline receptor antagonist
 -  Rilmenidine
 -  Moxonidine

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A. Vasodilator:

Arterioles only:hydralazine (Lupus erythematosus syndrome)

..minoxidil ...diazoxide??

Arterioles&venous:Na nitroprusside(CN toxicity)

Fenoldpam :D1 agonist

Hibiscus tea??

HTN Urgency	BP > 180/120 without organ dysfunction
HTN emergency	BP > 180/120 with organ dysfunction

➔ α .blocker

● Non selective: phentolamine

● Selective α_1 : prazosin..terazosin..Doxazocin

S/E: syncope like what??... Orthostatic hypotension

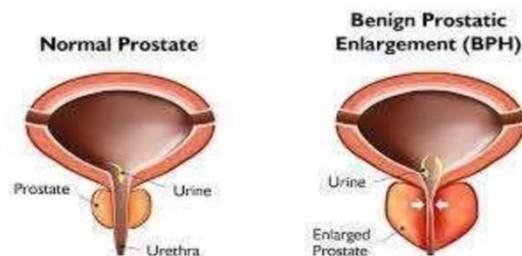
(BpH) Selective α_1 A:::

Tamsulin ,,,,,, Alfazocin (BpH),,,, silodosin).

Mechanism of action :

Tamsulosin is a selective α_1 receptor antagonist that has preferential selectivity for the α_{1a} receptor in the prostate versus the α_{1b} receptor in the blood vessels.

When alpha 1 receptors in the bladder neck and the prostate are blocked ,this causes a relaxation in the smooth muscle and therefore less resistance to urinary flow .due to this ,the pain associated with BPH can be reduce



Case?

A.W is an 85 year old man who presents to his physician with LUTS. A digital rectal examination confirm diagnosis of BPH , prostate volume is 31 gm ..pt suffer from orthostatic hypotention ..?

- A) terazocin
- b) finastride
- c) tamsulosin
- D) finastride +tamsulosin

Answer

Tamsulosin More selective ..less orthostatic hypotention
But finastride α reductase inhibitor .. if size greater than 40 gm

Worse BPH :

- α agonist (decongestant)
- anticholinergic(TCA,...)
- diuretics
- testosterone therapy

Treatment:

1. $\alpha 1$ - receptor antagonist: {Prazosin, Terazosin, Tamsulosin} # SE: orthostatic hypotension, nasal congestion, headache, floppy iris syndrome (with tamsulosin)
 2. 5 - α - reductase inhibitors: {Finasteride, Dutasteride} # used ONLY in prostate enlargement > 40 g
 3. Combination therapy: {tamsulosin + Finasteride or Dutasteride} # Symptoms of BPH with enlargement prostate > 40 g
 4. Phosphodiesterase – 5 inhibitors (PED-5): {ONLY tadalafil approved for BPH}
- Plant used in BPH: Saw palmetto

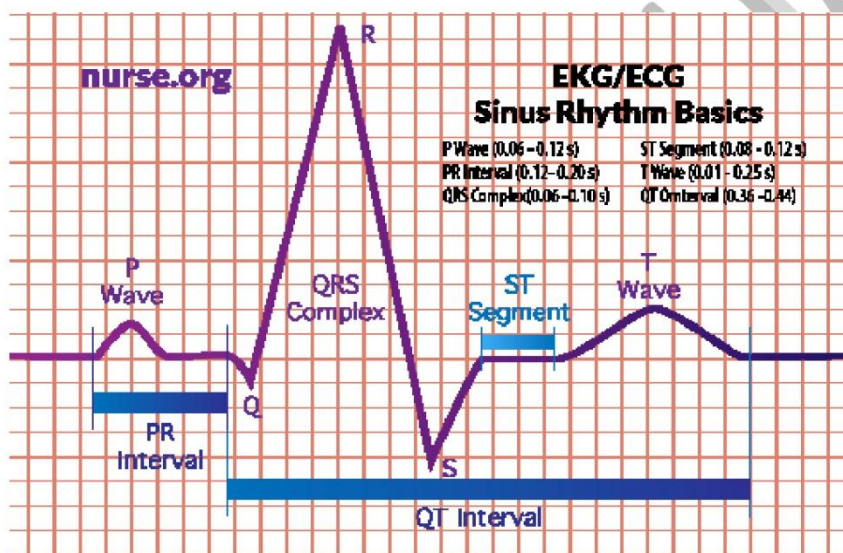
UTI  Cranberry

Separation ?

Tadalafil...Sildenafil ...vardenfil **VS** α blocker

➡ Shock

- Septic shock: 1st IV fluid 2nd NE or fixed doses of vasopressin.
- Septic shock with kidney injury: dopamine
- Cardiac shock: dopamine
- Anaphylactic shock: epinephrine (adrenaline) & ..&...

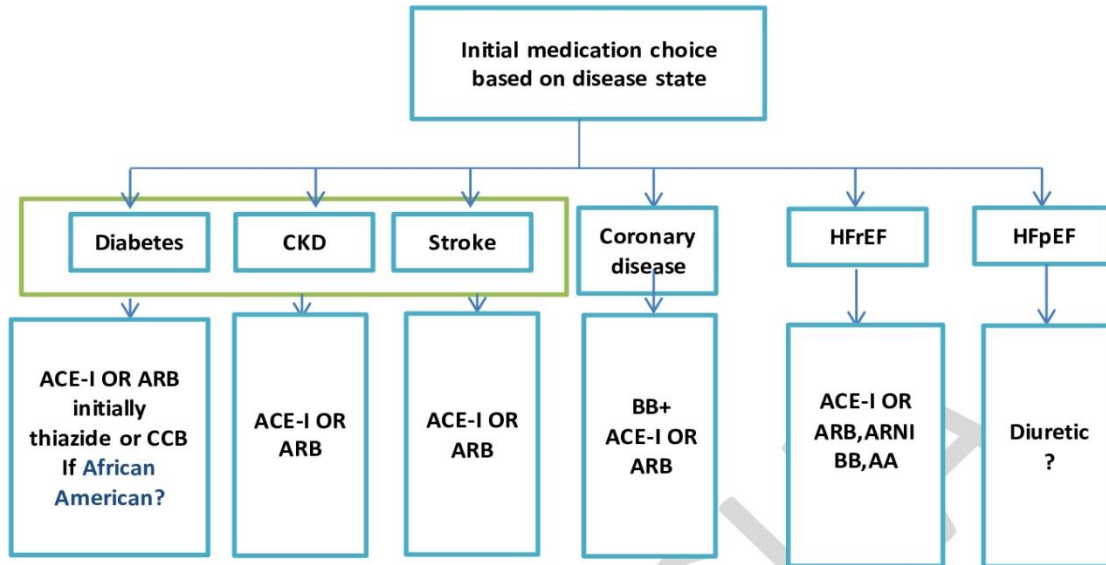


Absolute refractory period >>> QRS

Relative refractory period >>> ??

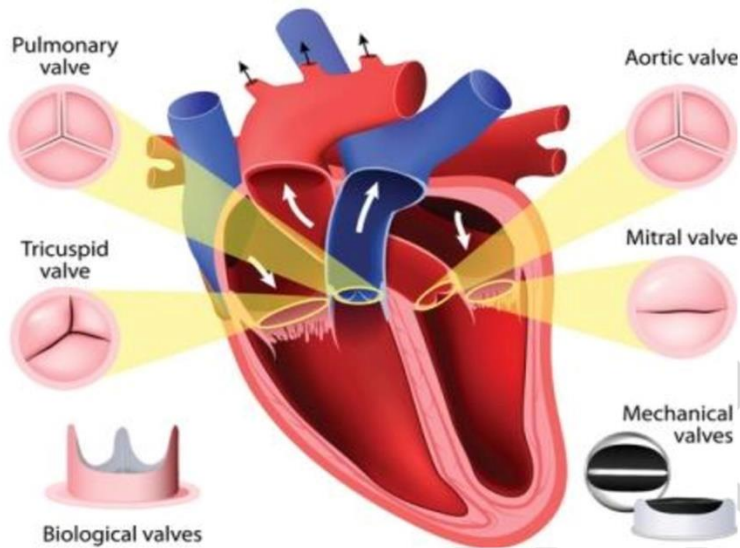
- **P wave:** Atrium depolarization
 - **QRS complex:** Ventricular depolarization
 - **T wave:** Ventricular Repolarization
 - **QT:** ventricular action potential
- PR?

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Heart valve



Arrhythmias

lub-dub" sounds`

- The first heart sound (S1) signals the **beginning** of ventricular systole (systole means to contract)
- The second heart sound (S2) signals the **end** of ventricular systole.
- Heart pump 5-6 liter every minute. HR 60-100

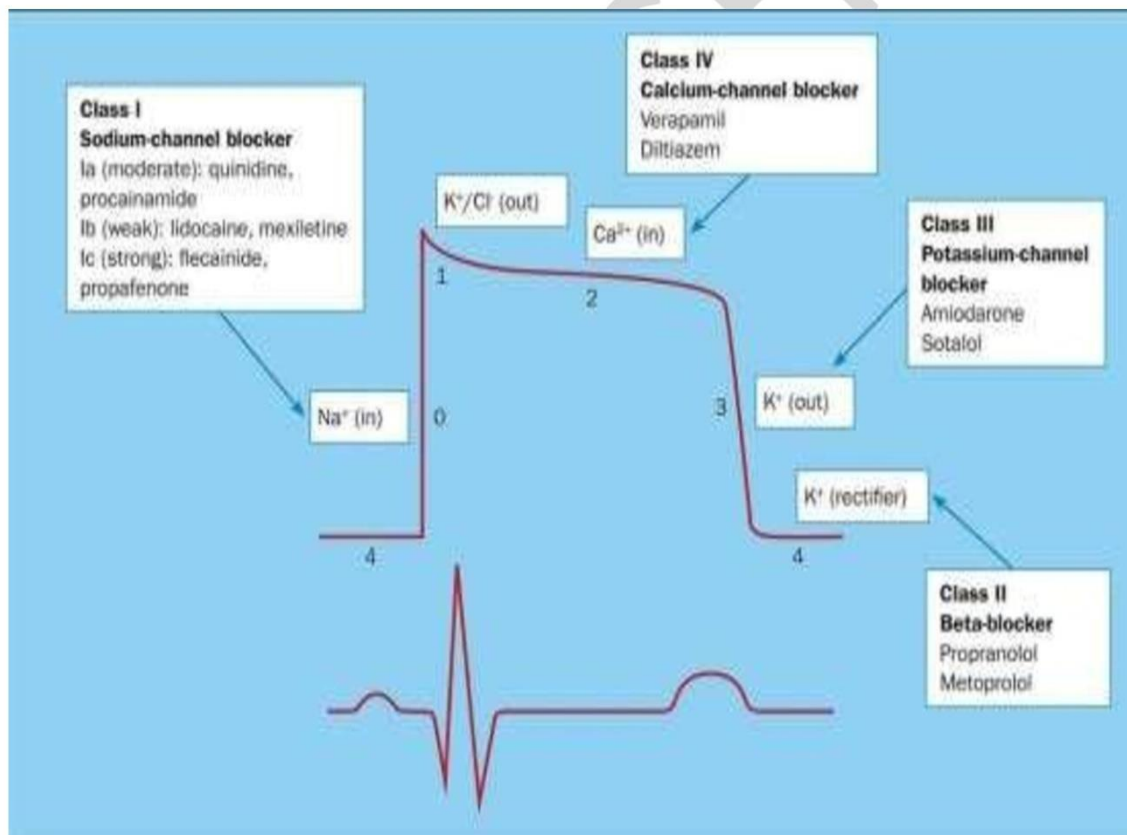
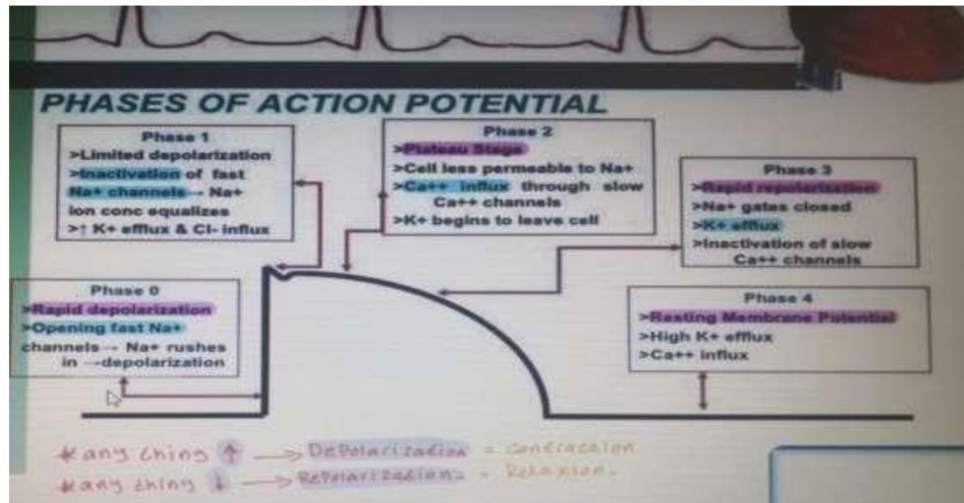
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Electrical signaling:

The Cardiac Action Potential

Phase 0	Rapid Ventricular depolarization due to influx of Na
Phase 1	Early rapid repolarization due to Na channel close
Phase 2	Plateau phase due to a Ca influx
Phase 3	Rapid Ventricular repolarization due to efflux of K
Phase 4	RMP, Atrial depolarization

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Classification of drug	Mechanism of action	Comment
IA	Na ⁺ channel blocker	Slows phase 0 depolarization in ventricular muscle fibers
IB	Na ⁺ channel blocker	Shortens phase 3 repolarization in ventricular muscle fibers
IC	Na ⁺ channel blocker	Markedly slows phase 0 depolarization b ventricular muscle fibers
II	B – Adrenoreceptor blocker	Inhibits phase 4 depolarization in SA & AV nodes
III	K ⁺ channel blocker	Prolongs phase 3 repolarization in ventricular muscle fibers
IV	Ca ⁺ channel blocker	Inhibits action potential in SA & AV nodes

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Classes

nbkc

A. Class I: Na⁺ Channel blockers –

Ia: Quinidine, Procainamide, Disopyramide

↓ ↓ ↓

SE Cinchonism
/Quinism, lupus erythematous, Anticholinergic effects
(blurred vision, tinnitus, ..) –

Ib: Lidocaine, Mexiletine, Phenyton

Ic: Flecainide, Propafenone (bronchospasm)

B. Class II: Beta-blockers. -ve inotropic

AVOID: with intrinsic sympathetic activity e.g (Pindolol, acebutolol)

C. Class III: K⁺ channel blockers - Amiodarone, Dronedarone, Sotalol, Dofetilide

Amiodarone SE: thyroid abnormalities, Blue-gray man syndrome, photosensitive, liver & pulmonary toxic

Sotalol is the ONLY one who have Beta blocker activity with K⁺ channel activity

D. Class IV: CCBs.

verapamil inhibits

metabolism of digoxin - Verapamil, Diltiazem

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Other classes

- Digoxin
- Adenosine DOC acute supraventricular tachy cardia
- Mg sulfate

Rate Control BB , CCB(non DHP) , digoxin

**Rhythm control Amiodarone , sotalol(class III) , propafenone,
flecainide(IC)..also (Ia)**

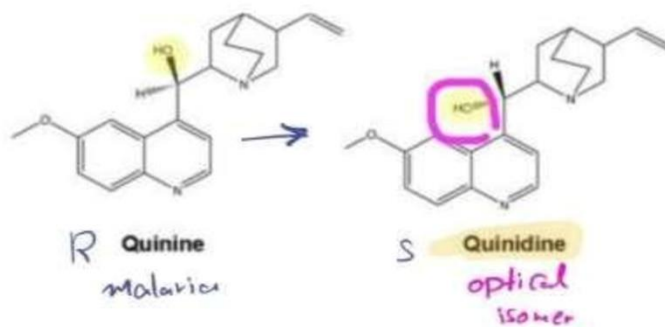
Rate & rhythm : sotalol ,amiodarone

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- A. procainamide : **Ventricular arrhythmia:**
- B. Arrhythmia with heart block: **phenytoin not CCB**
- C. Digitalis arrhythmia:**lidocaine or phenytoin.mg sulphate**
- D. Atrial fibrillation + HF:**class III (amiodarone) but can't use IC**
- E. Atrial fibrillation: **BB,,,CCB**
Arrhythmia Post MI : **amiodarone**

IC avoid in pt with HF or after MI

when quinine is configured

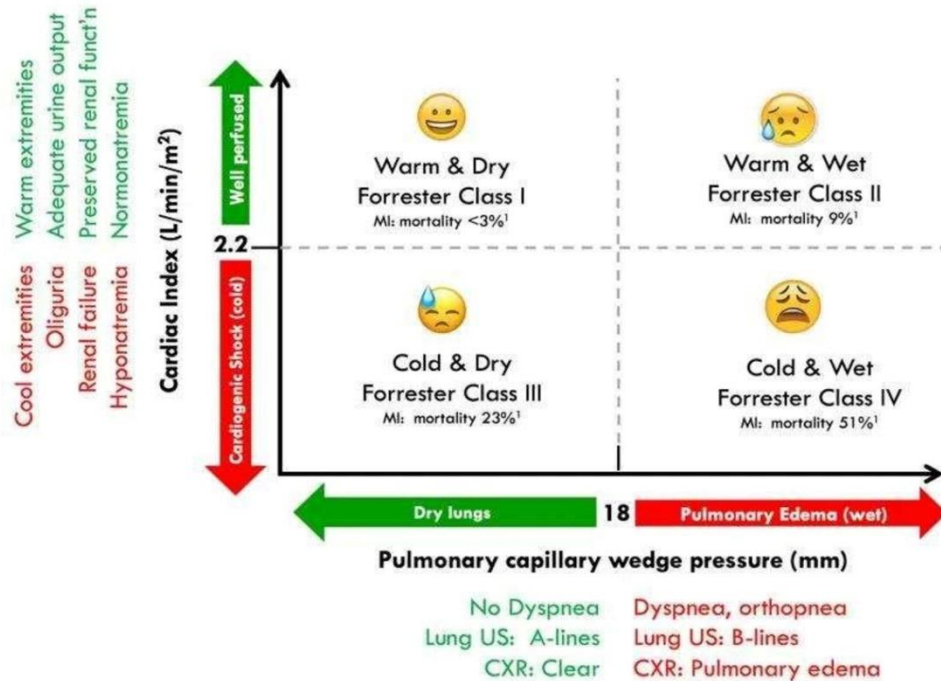


➤ Heart Failure:

- ★ **ACE-I and Beta blocker** should be given to ALL heart failure patient unless if there is contraindications to decrease mortality.
- ★ **Beta blocker in Heart failure (MBC): Metoprolol, Bisoprolol, Carvedilol # Diuretic in HF patient: Loop Diuretics**
- ★ **HF with EF < 40 which med. AVOID: CCBs**
- ★ **Drug Decrease mortality with HF: BBs, ACE-I, Spironolactone**
- ★ **Drug Increase mortality with HF: CCBs, Pioglitazone, gabapentin**



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(1) Mortality numbers from Forrester 1976 PMID 790191. Mortality is probably lower today.

For internet book of critical care, by @pulmccrit

Perfusion → warm or cold

Congestion or edema → wet or dry

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NYHA

	severity	Symptoms	exercise
1	Mild	✗	
2	Mild	✓	With moderate exercise
3	Moderate	✓	With minimal exercise
4	severe	✓	With no exercise (At rest)

NYHA Class



No limitation with ordinary physical activity

Slight limitation with ordinary physical activity results in breathlessness , fatigue , palpitation

marked limitation with less than ordinary physical activity results in breathlessness , fatigue

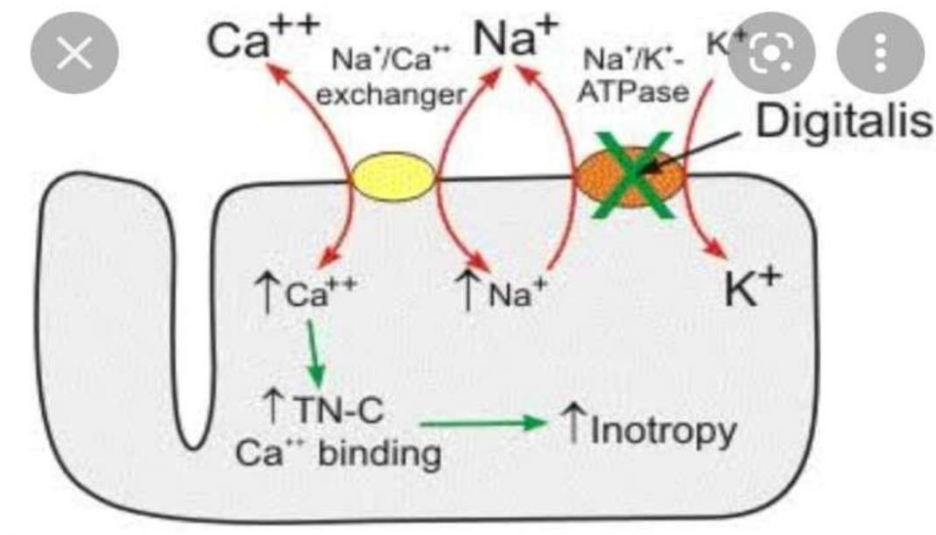
Symptoms at rest

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➤ **Heart Failure:**

Inotropic agent :

- a) **Cardiac glycoside:** digoxin ,digitoxin ,quabin **(chronic)**
- b) **B agonist:** dopamine & dobutamine **(acute)**
- c) **Phosphodiesterase III inhibitor :** amrinon, milrinon **(acute)**



Digoxin

MOA: inhibits the Na/K ATPase pump which results in a **positive inotropic** effect (increase in CO). It also exerts a parasympathetic effect which provides a **negative chronotropic effect** (decrease HR).

Dose: (lower dose for renal insufficiency, smaller, older, female) with the majority of patients taking no more than **0.125 mg daily**. Serum digoxin concentrations for HF should be kept < 1 ng/ml

➔ **Major side effect : cardiac arrhythmia ?? ttt??**
hyperkalemia

Factor increase **digitalis toxicity** : remember Li toxicity?

- **Hypokalemia (thiazide & loop diuretic)**
- **Renal failure (digoxin) , liver failure (digitoxin)**
- **Electrolyt disturbance : hypo k ,Mg and hyper ca**

Antidote : digibind,, digoxin immune fab

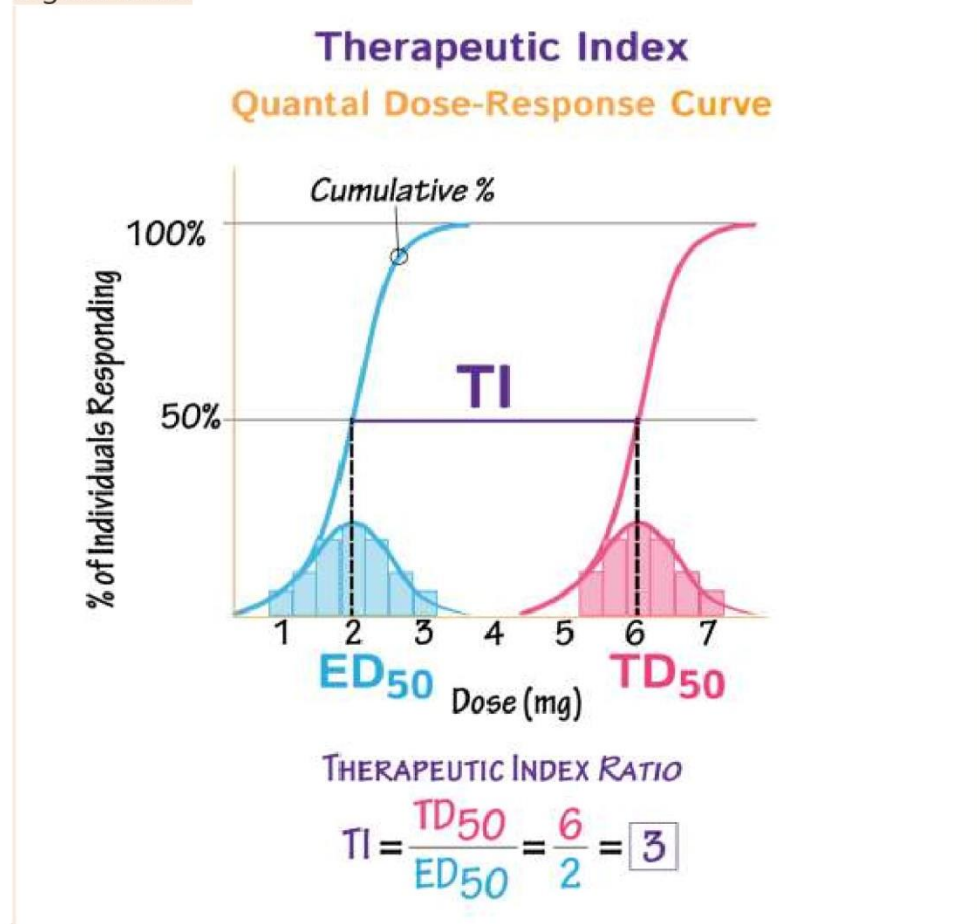
➔ **Digoxin 10-15 ng/ml or k > 5 meq/l**

Digitalis narrow therapeutic index

Test cardiac glycoside : killer killani

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Monitoring: ECG, HR, BP, **electrolytes**, renal function and digoxin level



Which of the following is the most appropriate time to draw a plasma blood sample for digoxin monitoring?

- A) 2 hours post-dose
- B) 4 hours post-dose
- C) 5 hours post-dose
- D) 6 hours post-dose

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Angiotensin receptor and neprilysin inhibitor (**ARNI**)

Sacubitril / Valsartan

Do not use with an ACE inhibitor. Must have **36**

hour wash-out period between stopping an ACE inhibitor and starting sacubitril/valsartan

DRUGS

➤ Angina:

Imbalance between O₂ demand & blood supply

High O₂ demand due to ↑ H.R

Low blood supply plague in coronary artery

Types :

- Stable
- Unstable
- Prinzmetal

Drug:

Nitrates , BB,CCB ,aspirin ,statin

Nitroglycerin, Isosorbide mononitrate Cause: **orthostatic hypotension , cause tolerance**

Sublingual tablet Patch (12 hr)

Drug interaction: **PDE-5-I** (tadalafil.....)

CI in angina: **Vasopressin**

Prinzmetal angina: **nitroglycerin & CCBs**



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<i>Separation</i>	Aspirin & ibuprofen Tadalafil & α blocker ACE-I & ARNI
--------------------------	---

DR/KABSHA

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Q

- . One drug approved to reduce mortality in heart failure??

beta blocker

- Responsible for generate impulse

SA node

- Propranolol similar to which one ?

Nadolol

- Food interaction with amiodarone ?

Grapefruit

- Which following have vasoconstriction effect?

Alpha 1

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- **Which drug cause constipation?**

Verapamil

- **Test of IHD?**

troponins

- **Vasodilator side effect?**

Tachycardia

- **Any of these have more alpha activity**

Epinephrine

- **Drug Disease interaction?**

Celecoxib and cardiac toxicity

- **Cardiac enzyme used in diagnosis of heart attack**

troponin enzyme

- **The best OTC for heart failure disease with knee pain.**

paracetamol > is The BEST with the presence of heart disease



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- Drug accelerate HF?

Gabapentin

- Beta blocker for HF patient?

Bisoprolol

- Beta blocker intrinsic sympathomimetic?

Acebutolol and pindolol

- Decrease mortality in HF?

beta blocker

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- Side effects CCB First-degree atrioventricular block and constipation are common dose dependent side effects of ?**verapamil**

- Absolute contra-indication for use fibrinolytic stroke **within 3 month**

- which drug contraindicated in CHF ?

Pioglitazone

- medication cause long QT and bradycardia? **propranolo**

- Dopamine is used in cardiac shock as ?

Increase .force of contraction

- Nitroprusside in congestive heart failure patient is administered by ?

slow i.v. infusion

- the action of digoxin can reduced with ?

antacid and hyperthyroidism

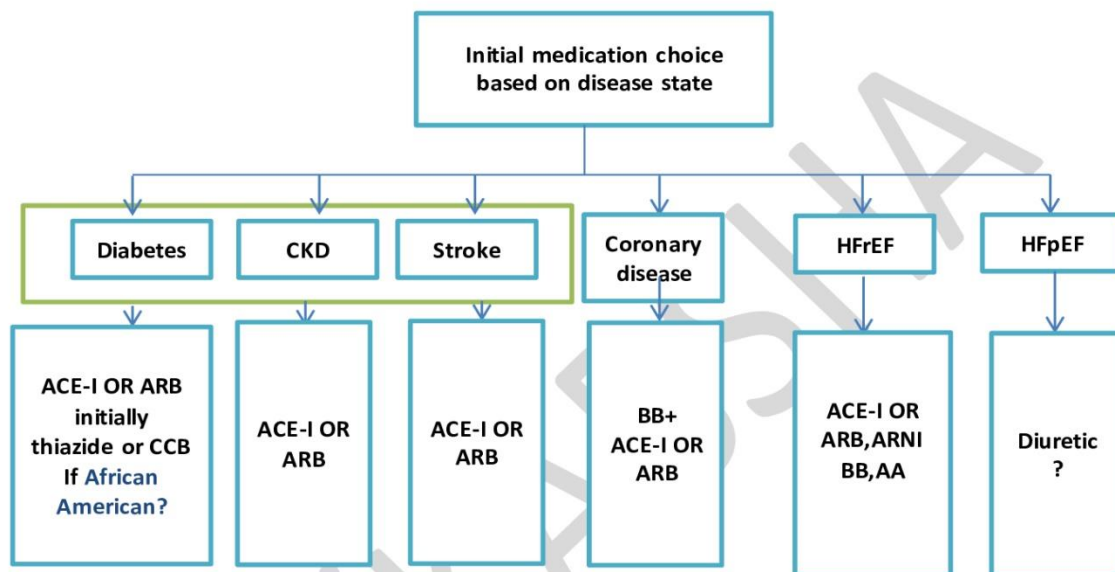
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- Hydralazine, ?
vessel relaxation. It dilates arterioles more than veins
- Food increase effect of ? Propranolol
 - Verapamil interaction with digoxin ?
 - phase 2 in cardiac mean that ?
- phase 3 in cardiac mean that ?
repolarization due to K⁺ outflux
 - Which of the following cardiac glycoside doesn't occur naturally?

Amrinone

- . Which medication considered as Rate control ?

Which medication considered as rhythm control ?



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1. hypersensitivity reaction to some drug ingestion

**four hours ago. Patient is given a drug which cause
dilation of vessels in muscle ,constriction of
cutaneous vessels ,and positive inotropic and
chronotropic effects on the heart Which of the
following is the most likely drug given**

A)Adrenaline

B)Isoproterenol

C)Acetylcholine

D)Metaproterenol

**2. 69-year-old woman presented with complains of
severe dry cough for a has week ,she has be recently
prescribed hydrochlorothiazide and captopril for
hypertension ,salbutamol and fexofenadine for
allergic asthma Which drug is the most likely reason
of the patient's complaint?**

A)Captopril

B)Salbutamol

C)Fexofenadine

D)Hydrochlorothiazide

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3. Which of the following is a consequence of stimulation of α 1-Adrenoceptors?

A. A flushing Adrenergic Receptors

B. increased heart rate

C. constricted bronchioles

D. elevated blood pressure

4. Which of the following is the mode of action of Phentolamine?

A. beta-adrenergic blocker

B. beta-adrenergic agonist

C. alpha-adrenergic agonist

D. alpha-adrenergic blocker (non selective alpha blocker)

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5. Which of the following drugs may decrease heart rate?

A. Hydrochlorothiazide

B. Isoproterenol

C. Amlodipine

D. Metoprolol

6. 69-year-old woman presented with complains of severe dry cough for a has week ,she has been recently prescribed hydrochlorothiazide and captopril for hypertension ,salbutamol and fexofenadine for allergic asthma Which drug is the most likely reason of thepatient's complaint?

A. Captopril

B. Salbutamol

C. Fexofenadine

D. Hydrochlorothiazide

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7. Which of the following medications is a class-IA

antiarrhythmic agent?

A. quinidine

B. flecainide

C. verapamil

D. amiodarone

8. Which one of the following conditions is a

contraindication for the use of enalapril?

A) pregnancy

B) hypertension

C) diabetes mellitus

D) congestive heart failure

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**9. which of the following is the best classification for
Bisoprolol?**

- A. beta 1 adrenergic receptor agonist**
- B. beta 1 adrenergic receptor blocker**
- C. alpha 1 adrenergic receptor blocker**
- D. non-selective beta-adrenergic receptor blocker**

**10. Which of the following is the Vaughan Williams
Classification for the antiarrhythmic drug
procainamide?**

- A) class IV**
- B) class III**
- C) class II**
- D) class IA**

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11. Which of the following can the pharmacological

property of hydralazine be directly responsible for?

- A) Relaxation of cardiac muscle
- B) Blockade of dopamine receptor
- C) Relaxation of vascular smooth muscle
- D) Contraction of vascular smooth muscle

12. A 31-year-old hypertensive man was treated for

addiction to amphetamine. He was also discharged on an antihypertensive drug. Later, he presented with severe depression. Which of the following medications was most likely prescribed?

- A) losartan
- B) captopril
- C) methyldopa
- D) hydrochlorothiazide

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**13.Which of the following is most likely the reason
for using amlodipine in the treatment of
hypertension?**

- A) selective beta 1 blocker**
- B) selective alpha 1 blocker**
- C) calcium channel blocker**
- D) selective serotonin reuptake inhibitor**

**14.year-old woman in her first trimester of
pregnancy who is recently diagnosed with
hypertension. Which of the following is the best and
safest therapy?**

- A. Aliskiren**
- B. Captopril**
- C. Valsartan**
- D. Labetalo**

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15. A 45-year-old lady known case hypertension

since one year on the following medications:

Lisinopril 5 mg orally once daily. Amlodipine 10 mg orally once daily. She was doing well and her blood pressure was controlled. In the last follow up visit, she informed her treating physician that she is planning to get pregnant. Which of the following is the most important decision to be taken for this patient before becoming pregnant?

- A. add methyldopa**
- B. discontinue lisinopril**
- C. continue same treatment in pregnancy**
- D. decrease lisinopril dose to 2.5 mg once daily**



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16. A 45-year-old patient with atrial fibrillation is treated with amiodarone 400 mg/day and rivaroxaban 20mg/day with the evening meal. He has concern about his drugs routine monitoring. Which of the following would be the best recommended regarding amiodarone monitoring?

- A. Slit-lamp examination at baseline and annually**
- B. Renal function tests at baseline and annually**
- C. Cardiac enzymes at baseline and annually**
- D. Liver function tests at baseline and every 6 months**

17.Which of the following beta-blockers is indicated for use in heart failure patients?

- A. Esmolol**
- B. Atenolol**
- C. Labetalol**
- D. Carvedilo**

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18.Which of the following medication should be prescribed to the patients with angina to treat an acute attack?

- A. Ranolazine**
- B. Nitroglycerin skin patch**
- C. Isosorbide dinitrate SR capsules**
- D. Nitroglycerin sublingual tablets**

19.Which of the following antihypertensive agents is most likely to cause peripheral edema?

- A. Atenolol**
- B. Perindopril**
- C. Amlodipine**
- D. Candesartan**

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20.Which of the following medications is a class-IC

antiarrhythmic agent?

A. Quinidine

B. Flecainide

C. Verapamil

D. Amiodarone

21.Which of the following is the pharmacological

property of atenolol?

A. nonselective beta-blocker

B. cardioselective beta 1-blocker

C. cardioselective beta 2-blocker

D. cardioselective beta 1-agonist

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22. What is the pharmacological property of cardiac

glycosides?

- A. alpha 1 blocker**
- B. potent bronchodilator**
- C. positive inotropic effect**
- D. negative inotropic effect.**

23.Which of the following is the reason of using

verapamil as an antiarrhythmic drug?

- A. it suppresses phase four depolarization**
- B. it stimulates the breakdown of adrenaline**
- C. it inhibits slow inward current of phase two of action potential**
- D. it increases the excitability of pyramidal neurons in the amygdala**

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24. Which one of the following conditions is a contraindication for the use of angiotensin converting enzyme inhibitors?

- A. hypertension**
- B. diabetes melitus**
- C. idiopathic angioedema**
- D. congestive heart failure**

25. A 75-year-old woman with congestive heart failure is prescribed digoxin to improve cardiac muscle contractility. She has a marked improvement in her symptoms. What cellular action of digoxin is responsible for its pharmacological action?

- A. Inhibition of cAMP synthesis**
- B. Inhibition of Na⁺ / K⁺ + ATPase enzyme**
- C. Inhibition of B adrenergic stimulation**
- D. Inhibition of mitochondrial calcium release**

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26.What is the main advantage of losartan over

Lisinopril?

- A. Less hyperkalemia**
- B. Safe during pregnancy**
- C. Less induction of cough**
- D. Higher oral bioavailability**

**27.How much the heart pump a liter of blood per
minute?**

- A. 0.5 Liters**
- B. 5 litres**
- C. 50 liters**

28.blood pressure in neonate ?

- A. same adult**
- B. Higher than adult**
- C. Lowed that adult**

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**29.Which of the following is a potential side effect
of chlorothiazide?**

- A. Hyperkalemia**
- B. Hypocalcemia**
- C. Hypouricemia**
- D. Hypokalemia**

**30.A man patient is placed on a new medication to
control his elevated BP. After one month he noted
that his breasts have become enlarged and tender.
Which of the following medication is the most
properly prescribed?**

- A. acetazolamide**
- B. chlorthalodone**
- C. spironolactone**
- D. hydrochlorothiazide**

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31.A 75-year-old man with NYHA-IV heart failure has been started on furosemide 120 mg intravenous infusion to improve pulmonary edema. Which of the following is the recommended duration time for furosemide infusion?

- A. 10 minutes**
- B. 20 minutes**
- C. 30 minutes**
- D. 40 minute**

32.When the patient at rest the volume of the blood found more in ?

- A- vein**
- B- arteria**

